

BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet

Microscopy

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. Write the magnification formula.
2. Define resolution.
3. Convert 2.5 mm to μm and to nm.

B Knowledge

4. How do you calculate total magnification of a light microscope?
5. Give two differences between a TEM and an SEM.
6. Why can a living amoeba be viewed with a light microscope but not with an electron microscope?

C Calculations

7. An image is 18 mm wide and magnification is $\times 600$. Calculate actual size in μm .
8. A $7.5 \mu\text{m}$ cell is drawn 3.0 cm wide. Calculate magnification.
9. A $10 \mu\text{m}$ scale bar measures 40 mm on a photograph. Calculate magnification.

D Exam-style practice

10. Calculate the actual width of a cell if the image is 35 mm wide at $\times 500$. Give the answer in μm . [2]
11. Compare the resolution of a light microscope and a TEM. [4]
12. Give two reasons for choosing a light microscope rather than an electron microscope. [2]

E Challenge

13. A student increases magnification from $\times 100$ to $\times 1000$ but cannot separate two close granules. Explain, using resolution.